HIDDEN EPOCH ARCHIVE

SPECIAL COLLECTIONS DIVISION

CLASSIFIED

# THE ANTIKYTHERA MECHANISM ANALYSIS

Recovered off Antikythera, Greece. 35.8617 N, 23.3067 E

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| CASE NO.: | HE-131-AK |
| FILE OPENED: | 1901 |
| STATUS: | UNRESOLVED - ACTIVE |
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## CASE SUMMARY

In 1901, sponge divers pulled a corroded lump of bronze out of a Roman-era shipwreck. It sat in a museum crate while archaeologists catalogued the statues. When the lump split open, it showed gear teeth. Precision-cut bronze gearwheels, from a ship that sank around 60 BC.

The Antikythera Mechanism is a geared analog computer. It modeled the sun, the moon, the phases, eclipses decades in advance, and, per the latest reconstructions, the five known planets, in a bronze case the size of a shoebox. Nothing of comparable mechanical complexity appears in the surviving record for the next fourteen hundred years.

The question of record was stated by the man who studied it longest, Derek de Solla Price: finding this device in that wreck is like finding a jet plane in the tomb of Tutankhamun. One does not build a computer first. One builds it last, at the end of a long line of simpler machines. The line is missing.

## TIMELINE OF RECORD

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| **c. 60 BC** | A cargo ship out of the eastern Aegean goes down off Antikythera carrying bronzes, glassware, coin and one geared instrument. |
| **1900** | Sponge divers from Symi, blown off course, find the wreck at 45 meters. Captain Kontos reports a seabed of corpses, the statues, to authorities in Athens. |
| **1901** | Salvage season recovers the hoard. One diver dies, two are paralyzed. Among the finds, an unremarkable corroded lump, crated. FILE OPENED. |
| **1902** | Archaeologist Valerios Stais notices a gear wheel embedded in the split lump. The find is disputed immediately; the objection on record is that the object is too complex for the wreck date. |
| **1951-1974** | Derek de Solla Price studies the fragments for two decades, publishes Gears from the Greeks, establishing at least 30 gears and the differential arrangement, a design not otherwise seen until the modern era. |

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| **2005** | X-ray microtomography reads the internal inscriptions, a built-in user manual naming eclipse cycles and games years. Tooth counts confirm the Metonic and Saros cycles in bronze. |
| **2021** | The UCL Antikythera Research Team publishes a full front-dial reconstruction with epicyclic planetary gearing. The count runs to some 69 gears in the complete device. |
| **Present** | Roughly a third of the mechanism survives. No second device, no workshop, no prototype and no fragment of a predecessor has ever been found. |


Fragment A, the largest surviving piece, National Archaeological Museum, Athens. The four-spoke main drive wheel is visible through the corrosion, along with the sandwiched gear plates behind it. This is what two thousand years of seawater leaves of a computer. The sea kept this unit out of the melting pot. Nothing kept the others.

Source: Photograph by Giovanni Dall'Orto, 2009. National Archaeological Museum, Athens. Used with attribution per lice

## THE OFFICIAL STORY

The consensus attributes the mechanism to Hellenistic craftsmen, probably of the school of Hipparchus or later Rhodes workshops, second or first century BC. Cicero mentions geared planetaria associated with Archimedes and Posidonius, so the capability is said to be documented. The device is called the culmination of a Greek mechanical tradition.

A culmination with no line to culminate. The tradition that supposedly produced it left no other gearwork of remotely comparable complexity, no workshop debris, no apprentice pieces, no failed castings, nothing. The literary references are one-line remarks about wonders owned by famous men, exactly the shape a rumor takes. The physical record contains this one object, alone.

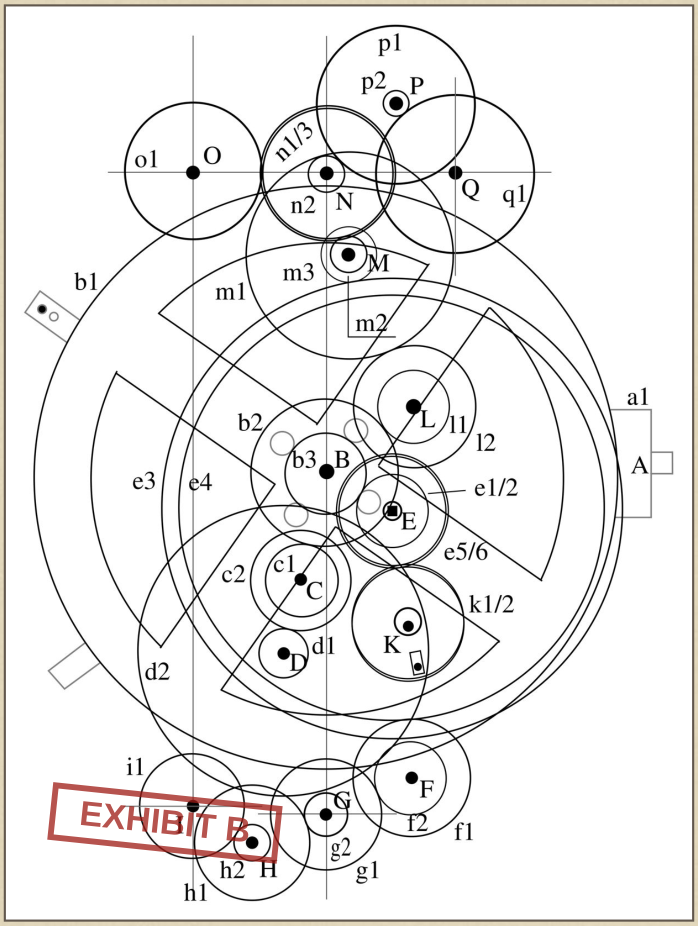
## WHAT THE RECORD SHOWS

The engineering, first. The surviving gears are cut from bronze sheet around two millimeters thick, teeth triangular, some wheels carrying 223 teeth, a prime number chosen because the Saros eclipse cycle is 223 lunar months. A pin-and-slot arrangement rides one wheel on another to reproduce the moon's variable speed, which means the maker mechanically encoded lunar anomaly, an effect not formally theorized until Kepler. The tooth counts are not decoration. They are astronomy, compiled.

The device knew things its era is not supposed to have known this precisely. The inscriptions carry eclipse predictions with times of day and color forecasts for the eclipsed moon. The 2021 reconstruction gives the planets their correct synodic periods through compound gear trains accurate to a day in decades. This is not a model in the display sense. It is computation, executed in metal.

Second, the isolation. Price wrote that the mechanism forces a complete rethink of Greek technology, and then spent twenty years watching the field decline to rethink it. The standard histories still date geared clockwork to medieval Europe. The one surviving counterexample is handled as a marvel, an outlier, a [redacted] vanished]] except for the unit that happened to sink in deep water where no one could melt it down.

That last clause is the uncomfortable one. Bronze was money. Every geared device that stayed on land was eventually scrap. The only reason this file exists is that the sea kept one unit out of reach. The correct inference from one survivor in an environment that destroys survivors is not that one was made. It is that the others did not survive. The wreck did not preserve an exception. It preserved the only sample.




Reconstructed gear-train schematic. Every axle and wheel shown here is attested in the surviving fragments or required by the tooth counts. Metonic dial, Saros eclipse dial, Exeligmos corrector, lunar anomaly pin-and-slot, all driven from one hand crank. This is the anatomy of the machine the textbooks call an isolated curiosity.

Source: Gear-train schematic released CC0 via Wikimedia Commons.

## ANOMALIES ON FILE

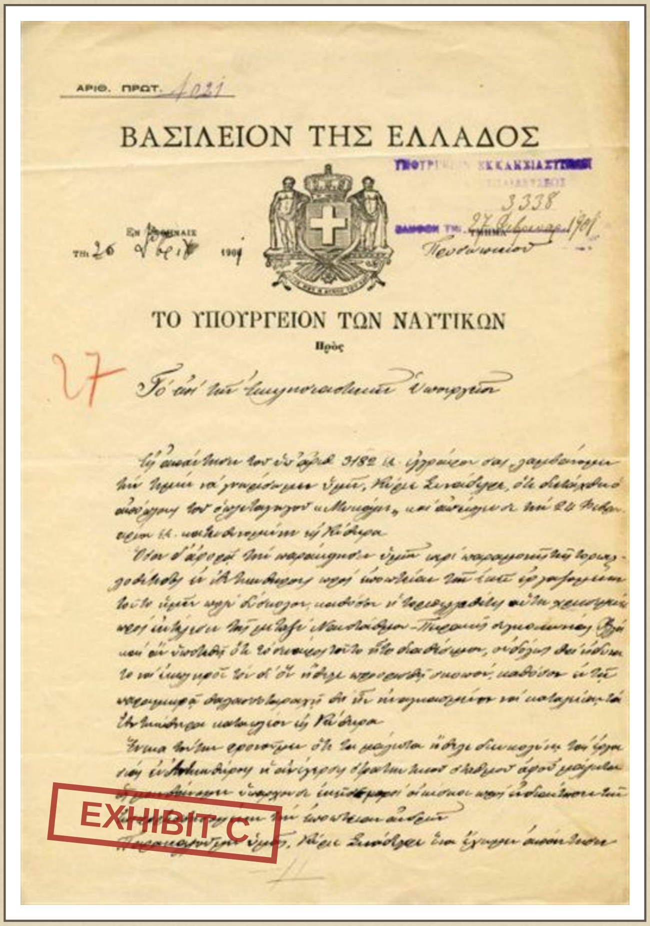
ITEM 1. Complexity without ancestry. Some 69 gears, differential and epicyclic arrangements, in a record that otherwise contains no Greek gearwork more advanced than a windlass. The developmental chain is one hundred percent missing.

ITEM 2. Complexity without descendants. The skills evaporate for fourteen centuries, until European astronomical clocks, which arrive, again, near fully formed, in the same corner of the world the wreck's cargo was sailing toward.

ITEM 3. The maker encoded lunar anomaly mechanically eighteen centuries before the mathematics existed to describe it formally.

ITEM 4. The initial 1902 identification was resisted on the explicit grounds that the object was too advanced for its date. The date won. The implications were then filed under craftsmanship and left there.

ITEM 5. Expeditions returned in 1953, 1976, 2012 and every season since 2014. The wreck is still producing material, and the excavators state on record that most of the cargo field is untouched. If a second mechanism exists anywhere, it is there.




Hellenic Ministry of Naval Affairs dispatch, 26 February 1901, ordering the vessel Mykali to the Antikythera wreck for the salvage season. The paperwork that pulled the mechanism off the seabed. The government filed the recovery. The field filed the implications, and never retrieved them.

Source: Hellenic Ministry of Naval Affairs, 1901. Public domain.

## FINAL ASSESSMENT

The Antikythera Mechanism is treated as a delightful exception, and exceptions are where the record admits what it otherwise conceals. A civilization does not produce one computer. The device testifies to a mature engineering lineage, generations deep, whose every other product was recycled into coins and cannon metal, and whose existence was compressed by history into two sentences of Cicero.

This office assesses the mechanism as recovered property of a lost technical tradition, scale unknown, duration unknown, and notes that the knowledge it embodied had to come from somewhere and go somewhere. Fourteen hundred years is not a gap. It is an erasure.

The cargo field at 45 meters is still being worked. The file holds position until the sea gives up the second unit.

CASE REMAINS OPEN